Project

2026-04-09

How are housing property characteristics and location associated with housing prices?

We study how housing characteristics and location affect housing prices using Bayesian regression models. The response variable is log-transformed to reduce skewness and stabilize variance.

Data cleaning

```r
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
data <- read.csv("Bengaluru_House_Price_Data(main).csv")
head(data)
```

```
##              area_type  availability                 location      size society
## 1 Super built-up  Area        19-Dec Electronic City Phase II     2 BHK Coomee
## 2           Plot  Area Ready To Move         Chikka Tirupathi 4 Bedroom Theanmp
## 3       Built-up  Area Ready To Move              Uttarahalli     3 BHK
## 4 Super built-up  Area Ready To Move       Lingadheeranahalli     3 BHK Soiewre
## 5 Super built-up  Area Ready To Move                 Kothanur     2 BHK
## 6 Super built-up  Area Ready To Move               Whitefield     2 BHK DuenaTa
##   total_sqft bath balcony  price
## 1       1056    2       1  39.07
## 2       2600    5       3 120.00
## 3       1440    2       3  62.00
## 4       1521    3       1  95.00
## 5       1200    2       1  51.00
## 6       1170    2       1  38.00
```

```r
# 1. Remove column with too many missing values
data$society <- NULL

# 2. Extract numeric bedrooms from size
data$bedrooms <- as.numeric(gsub("[^0-9]", "", data$size))

# 3. Clean total_sqft
convert_sqft <- function(x) {
  if (grepl("-", x)) {
    nums <- as.numeric(unlist(strsplit(x, "-")))
    return(mean(nums))
  } else if (grepl("^[0-9.]+$", x)) {
    return(as.numeric(x))
  } else {
    return(NA)
  }
}

data$sqft <- sapply(data$total_sqft, convert_sqft)

# 4. Transform balcony
data$balcony <- factor(data$balcony, levels = c(0,1,2,3))

# 5. Remove rows with missing key variables
data <- data[!is.na(data$sqft), ]
data <- data[!is.na(data$bath), ]
data <- data[!is.na(data$balcony), ]
data <- data[!is.na(data$bedrooms), ]
data <- data[!is.na(data$location), ]
data <- data[data$sqft > 0, ]

# 6. Frequency-based grouping for location
location_counts <- data %>%
  group_by(location) %>%
  tally(sort = TRUE)

top_18_names <- head(location_counts$location, 18)

data <- data %>%
  mutate(location_grouped = case_when(
    location %in% top_18_names ~ as.character(location),
    location_counts$n[match(location, location_counts$location)] >= 30 ~ "Other_Established",
    TRUE ~ "Other_Sparse"
  ))

# Convert to factor
all_levels <- c(as.character(top_18_names), "Other_Established", "Other_Sparse")
data$location_grouped <- factor(data$location_grouped, levels = all_levels)

# 7. Log-transform variables
data$log_price <- log(data$price)
data$log_sqft  <- log(data$sqft)
```

```r
# 8. Keep relevant variables
clean_data <- data[, c("log_price",
                       "log_sqft",
                       "balcony",
                       "location_grouped")]

# 9. Rename columns
colnames(clean_data) <- c("log_price",
                          "log_sqft",
                          "balcony",
                          "location")

# 10. Save cleaned dataset
write.csv(clean_data, "clean_housing.csv", row.names = FALSE)

# 11. Check data
head(clean_data)
```

```
##   log_price log_sqft balcony                 location
## 1  3.665355 6.962243       1 Electronic City Phase II
## 2  4.787492 7.863267       3             Other_Sparse
## 3  4.127134 7.272398       3              Uttarahalli
## 4  4.553877 7.327123       1             Other_Sparse
## 5  3.931826 7.090077       1        Other_Established
## 6  3.637586 7.064759       1               Whitefield
```

```r
table(clean_data$location)
```

```
##
##               Whitefield           Sarjapur  Road          Electronic City
##                      513                      372                      300
##           Kanakpura Road              Thanisandra                Yelahanka
##                      259                      230                      206
##              Uttarahalli                   Hebbal    Raja Rajeshwari Nagar
##                      186                      173                      168
##             Marathahalli       7th Phase JP Nagar        Bannerghatta Road
##                      164                      146                      144
##              Hennur Road             Haralur Road Electronic City Phase II
##                      142                      135                      130
##             Rajaji Nagar               Chandapura                Bellandur
##                       99                       98                       91
##        Other_Established             Other_Sparse
##                     3888                     5225
```

```r
library(rstan)

clean_data$balcony <- factor(clean_data$balcony, levels = c(0,1,2,3))

# Create 3 dummy variables (baseline = balcony 0)
Z <- model.matrix(~ balcony, data = clean_data)[, -1]

# The data for Stan
```

```r
stan_data <- list(
  N = nrow(clean_data),
  log_price = clean_data$log_price,
  log_sqft = clean_data$log_sqft,
  balcony_dummies = Z
)

# Fit model using Stan
fit <- stan(
  file = "model1.stan",
  data = stan_data,
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  refresh = 0
)
```

```
## Warning in readLines(file, warn = TRUE): incomplete final line found on
## '/Users/maidoufude/Desktop/model1.stan'
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.5)'
## using SDK: 'MacOSX15.5.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Li
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen
## /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/src/Cor
##   679 | #include <cmath>
##       |          ^~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

```r
# Posterior means
post <- rstan::extract(fit)

beta0_hat <- mean(post$beta0)
beta1_hat <- mean(post$beta1)
beta_balcony_hat <- colMeans(post$beta_balcony)
sigma_hat <- mean(post$sigma)

beta0_hat
```

```
## [1] -3.914453
```

```r
beta1_hat
```

```
## [1] 1.163446
```

```r
beta_balcony_hat
```

```
## [1] -0.14391914 -0.12892016 -0.04516948
```

```r
sigma_hat
```

```
## [1] 0.4654007
```

```r
# credible intervals
beta0_ci <- quantile(post$beta0, probs = c(0.025, 0.975))
beta1_ci <- quantile(post$beta1, probs = c(0.025, 0.975))
beta_balcony_ci <- apply(post$beta_balcony, 2, quantile, probs = c(0.025, 0.975))
sigma_ci <- quantile(post$sigma, probs = c(0.025, 0.975))

beta0_ci
```

```
##      2.5%     97.5%
## -4.054867 -3.778001
```

```r
beta1_ci
```

```
##     2.5%    97.5%
## 1.144638 1.182548
```

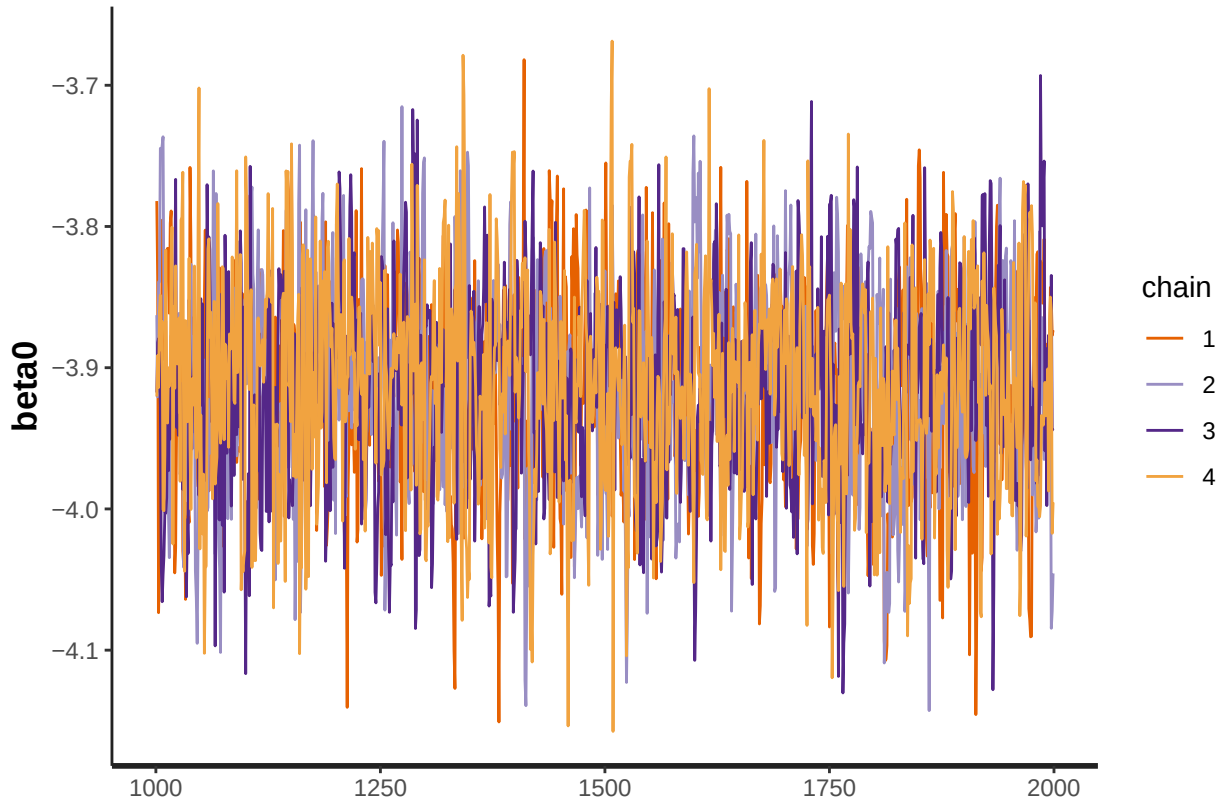
```r
beta_balcony_ci
```

```
##
##               [,1]       [,2]        [,3]
##   2.5%  -0.1758236 -0.1600130 -0.08130457
##   97.5% -0.1131398 -0.0982027 -0.00940855
```

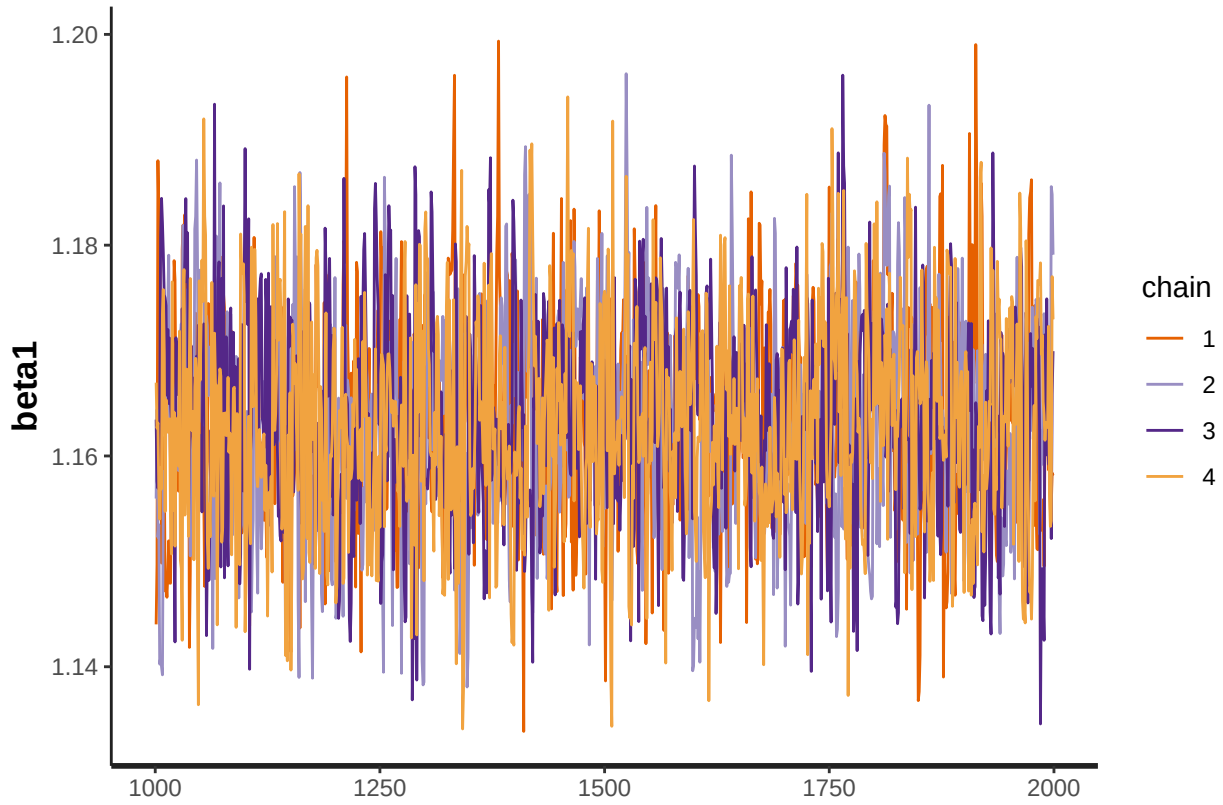
```r
sigma_ci
```

```
##      2.5%     97.5%
## 0.4597233 0.4710378
```

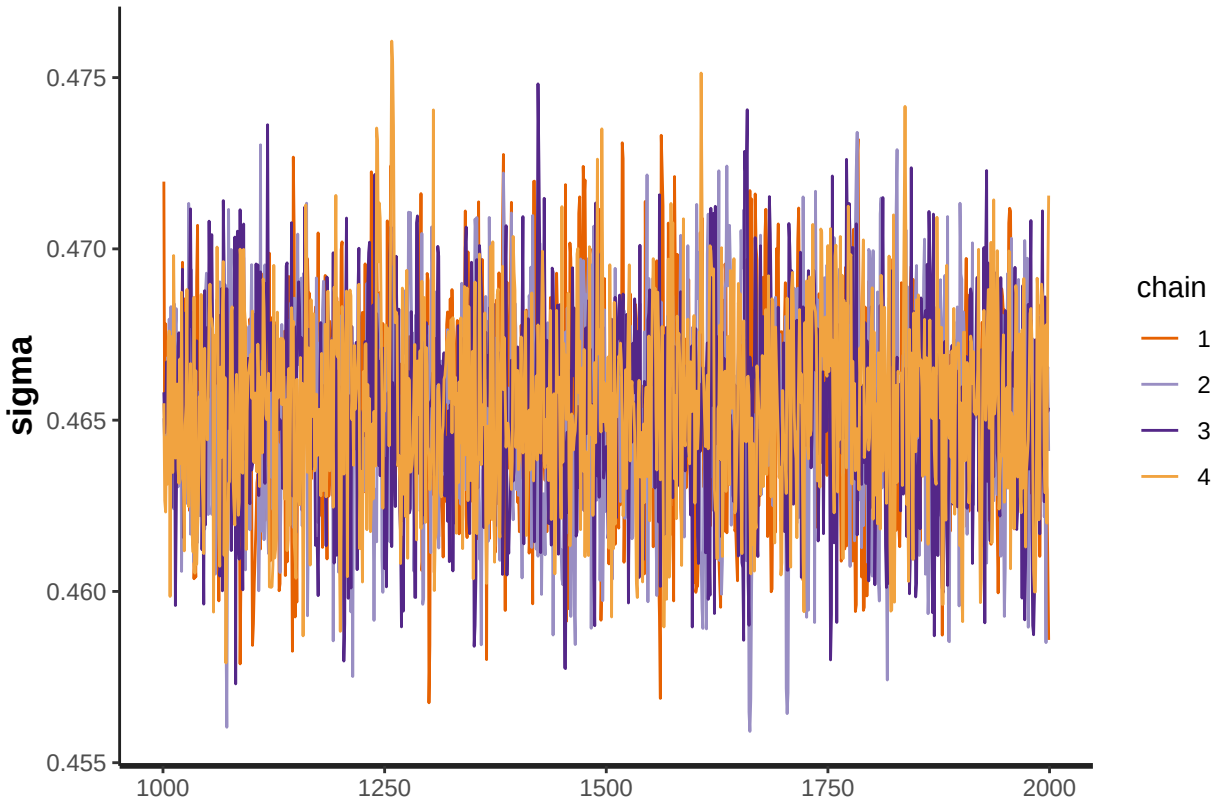
```r
traceplot(fit, pars = "beta0")
```



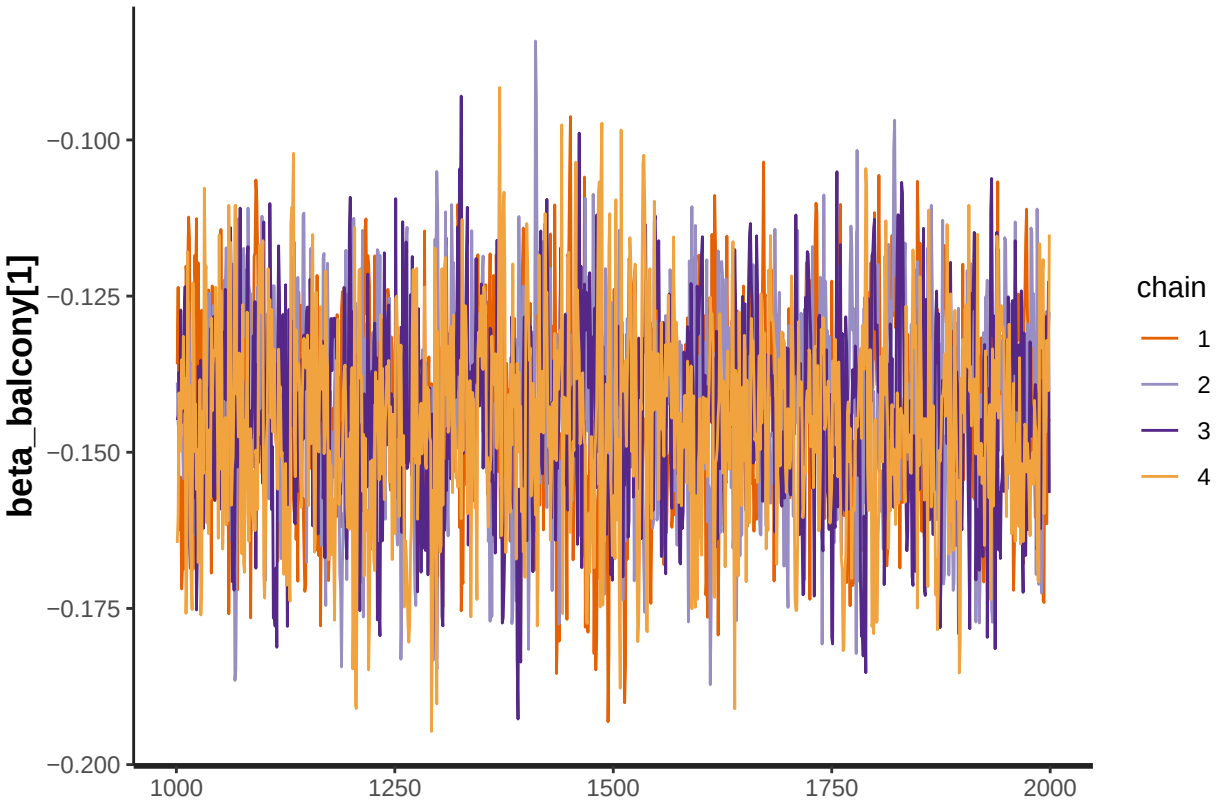

```
traceplot(fit, pars = "beta1")
```



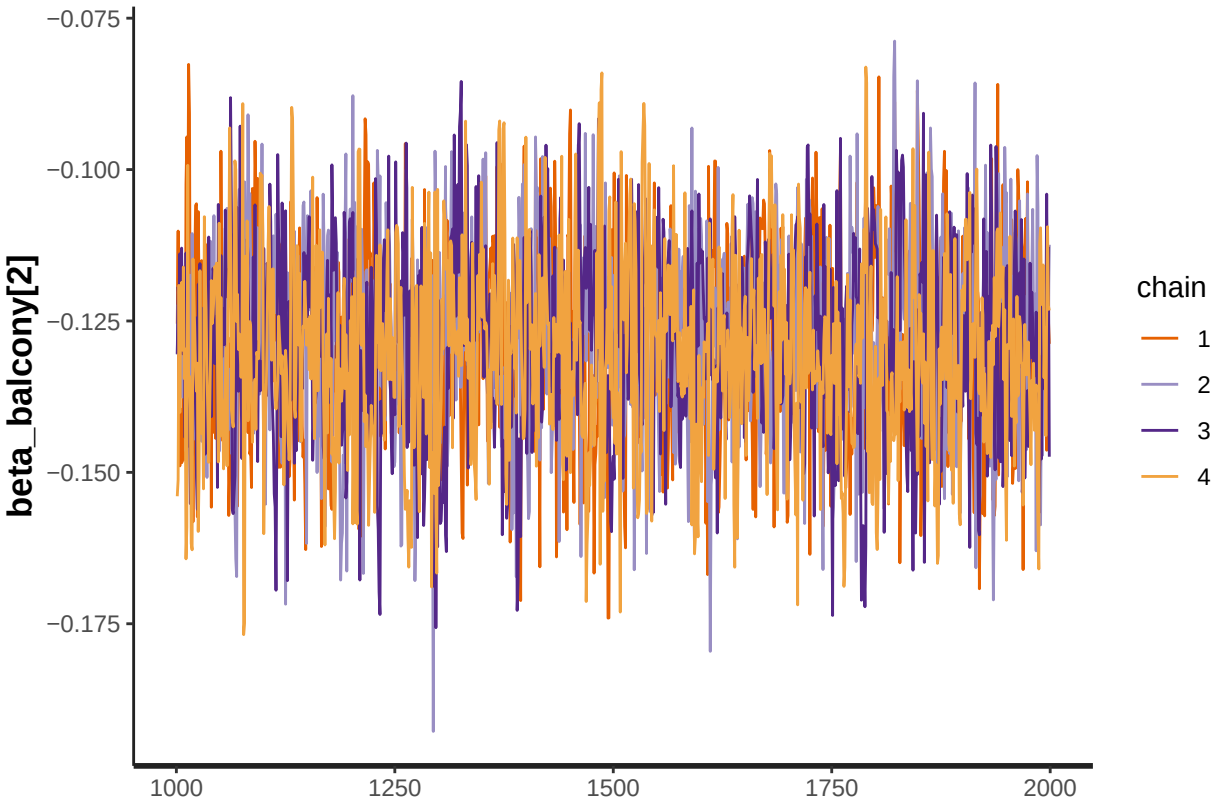

```r
traceplot(fit, pars = "sigma")
```



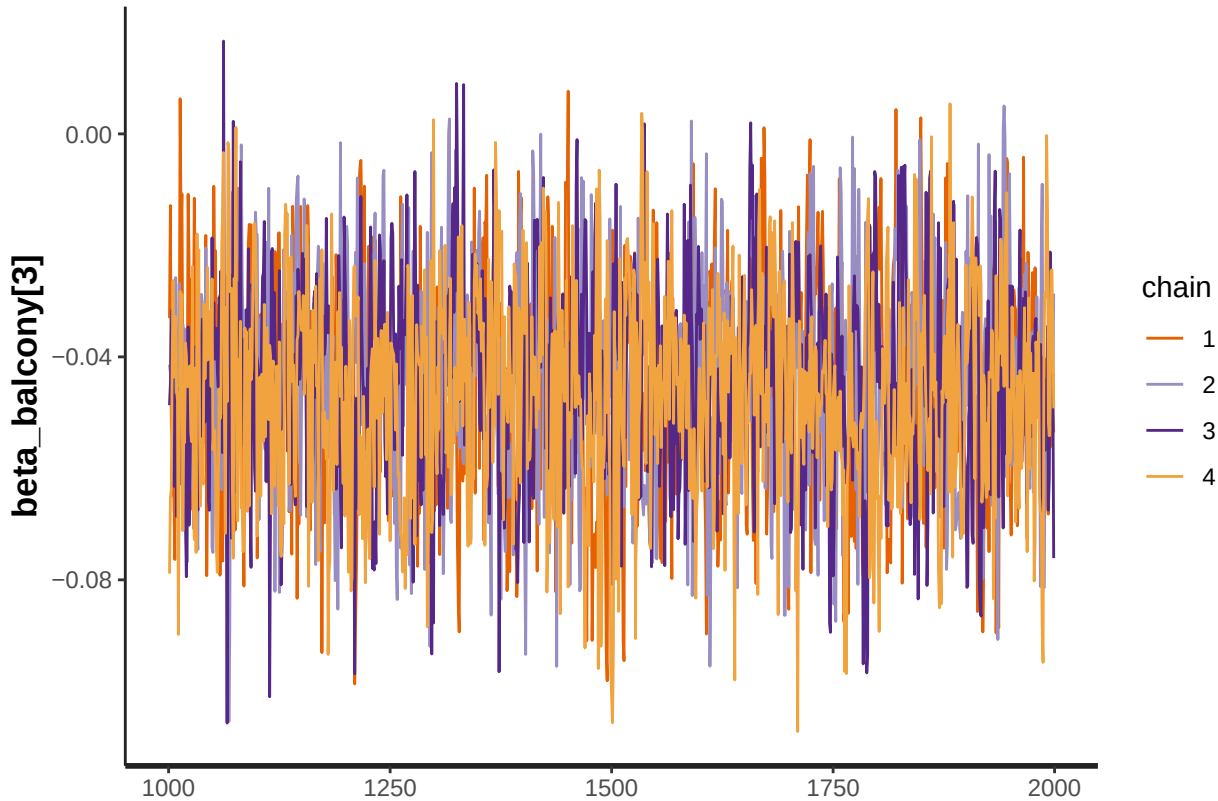

```
traceplot(fit, pars = "beta_balcony[1]")
```




```
traceplot(fit, pars = "beta_balcony[2]")
```




```
traceplot(fit, pars = "beta_balcony[3]")
```




```r
# summary diagnostics
diag_summary <- summary(
  fit,
  pars = c("beta0", "beta1", "beta_balcony", "sigma"),
  probs = c(0.025, 0.5, 0.975)
)$summary

diag_summary[, c("mean", "sd", "n_eff", "Rhat")]
```

```
##                        mean          sd    n_eff      Rhat
## beta0           -3.91445350 0.070488578 1628.996 0.9999451
## beta1            1.16344562 0.009705725 1674.747 0.9996805
## beta_balcony[1] -0.14391914 0.015736996 1610.719 1.0020388
## beta_balcony[2] -0.12892016 0.015716721 1662.714 1.0013678
## beta_balcony[3] -0.04516948 0.018272496 1711.180 1.0005224
## sigma            0.46540072 0.002890446 2528.248 0.9999653
```

```r
# Synthetic data validation

# known true parameters
beta0_true <- -3.90
beta1_true <- 1.16
beta_balcony_true <- c(-0.14, -0.13, -0.05)
sigma_true <- 0.47
```

```r
# simulate synthetic response
mu_syn <- beta0_true + beta1_true * clean_data$log_sqft + Z %*% beta_balcony_true
log_price_syn <- rnorm(nrow(clean_data), mean = mu_syn, sd = sigma_true)

stan_data_syn <- list(
  N = nrow(clean_data),
  log_price = as.vector(log_price_syn),
  log_sqft = clean_data$log_sqft,
  balcony_dummies = Z
)

fit_syn <- stan(
  file = "model1.stan",
  data = stan_data_syn,
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  refresh = 0
)
```

```
## Warning in readLines(file, warn = TRUE): incomplete final line found on
## '/Users/maidoufude/Desktop/model1.stan'
```

```r
post_syn <- rstan::extract(fit_syn)

# compare true values with posterior means
recovery_table <- data.frame(
  Parameter = c("beta0", "beta1", "beta_balcony1", "beta_balcony2", "beta_balcony3", "sigma"),
  True = c(beta0_true, beta1_true, beta_balcony_true, sigma_true),
  Estimate = c(
    mean(post_syn$beta0),
    mean(post_syn$beta1),
    colMeans(post_syn$beta_balcony),
    mean(post_syn$sigma)
  )
)

recovery_table
```

```
##       Parameter  True    Estimate
## 1         beta0 -3.90 -3.84073642
## 2         beta1  1.16  1.15200732
## 3 beta_balcony1 -0.14 -0.14185705
## 4 beta_balcony2 -0.13 -0.13163347
## 5 beta_balcony3 -0.05 -0.04506978
## 6         sigma  0.47  0.46734458
```

```r
library(cmdstanr)
```

```
## Warning: package 'cmdstanr' was built under R version 4.5.2
```

```r
library(posterior)
```

```
## Warning: package 'posterior' was built under R version 4.5.2
```

```r
#

mod <- cmdstan_model("model1.stan")
```

```
## Warning in readLines(stan_file): incomplete final line found on 'model1.stan'
```

```r
fit_vi <- mod$variational(
  data = stan_data,
  seed = 123,
  algorithm = "meanfield",
  output_samples = 2000,
  refresh = 0,
  show_messages = FALSE,
  show_exceptions = FALSE
)
```

```r
draws_df <- as_draws_df(fit_vi$draws())

beta0_vi <- mean(draws_df$beta0)
beta1_vi <- mean(draws_df$beta1)
sigma_vi <- mean(draws_df$sigma)

beta_balcony_vi <- c(
  mean(draws_df$`beta_balcony[1]`),
  mean(draws_df$`beta_balcony[2]`),
  mean(draws_df$`beta_balcony[3]`)
)

beta0_vi
```

```
## [1] -2.37597
```

```r
beta1_vi
```

```
## [1] 0.9626007
```

```r
beta_balcony_vi
```

```
## [1] -0.19965819 -0.13667826 -0.01218419
```

```r
sigma_vi
```

```
## [1] 0.4720208
```

```r
# credible intervals
beta0_vi_ci <- quantile(draws_df$beta0, probs = c(0.025, 0.975))
beta1_vi_ci <- quantile(draws_df$beta1, probs = c(0.025, 0.975))
sigma_vi_ci <- quantile(draws_df$sigma, probs = c(0.025, 0.975))

beta_balcony_vi_ci <- rbind(
  quantile(draws_df$`beta_balcony[1]`, probs = c(0.025, 0.975)),
  quantile(draws_df$`beta_balcony[2]`, probs = c(0.025, 0.975)),
  quantile(draws_df$`beta_balcony[3]`, probs = c(0.025, 0.975))
)
beta0_vi_ci
```

```
##      2.5%     97.5%
## -2.386455 -2.366059
```

```r
beta1_vi_ci
```

```
##      2.5%     97.5%
## 0.9608635 0.9643985
```

```r
sigma_vi_ci
```

```
##      2.5%     97.5%
## 0.4651750 0.4791312
```

```r
beta_balcony_vi_ci
```

```
##            2.5%       97.5%
## [1,] -0.2174652 -0.18255118
## [2,] -0.1509890 -0.12218520
## [3,] -0.0376877  0.01294812
```

```r
compare_table <- data.frame(
  Parameter = c("beta0", "beta1", "beta_balcony1", "beta_balcony2", "beta_balcony3", "sigma"),
  MCMC = c(beta0_hat, beta1_hat, beta_balcony_hat, sigma_hat),
  VI = c(beta0_vi, beta1_vi, beta_balcony_vi, sigma_vi)
)

compare_table
```

```
##       Parameter        MCMC          VI
## 1         beta0 -3.91445350 -2.37596986
## 2         beta1  1.16344562  0.96260073
## 3 beta_balcony1 -0.14391914 -0.19965819
## 4 beta_balcony2 -0.12892016 -0.13667826
## 5 beta_balcony3 -0.04516948 -0.01218419
## 6         sigma  0.46540072  0.47202082
```

```r
# Synthetic data validation (VI)

beta0_true_vi <- -2.38
beta1_true_vi <- 0.96
beta_balcony_true_vi <- c(-0.19, -0.13, -0.012)
sigma_true_vi <- 0.47

# simulate synthetic response
mu_syn_vi <- beta0_true_vi + beta1_true_vi * clean_data$log_sqft + Z %*% beta_balcony_true_vi
log_price_syn_vi <- rnorm(nrow(clean_data), mean = mu_syn_vi, sd = sigma_true_vi)

stan_data_syn_vi <- list(
  N = nrow(clean_data),
  log_price = as.vector(log_price_syn_vi),
  log_sqft = clean_data$log_sqft,
  balcony_dummies = Z
)

# fit VI on synthetic data
fit_vi_syn <- mod$variational(
  data = stan_data_syn_vi,
  seed = 123,
  algorithm = "meanfield",
  output_samples = 2000,
  refresh = 0,
  show_messages = FALSE,
  show_exceptions = FALSE
)
```

```r
draws_syn_vi <- as_draws_df(fit_vi_syn$draws())

# compare true values with posterior means
recovery_table_vi <- data.frame(
  Parameter = c("beta0", "beta1", "beta_balcony1", "beta_balcony2", "beta_balcony3", "sigma"),
  True_vi = c(beta0_true_vi, beta1_true_vi, beta_balcony_true_vi, sigma_true_vi),
  Estimate_vi = c(
    mean(draws_syn_vi$beta0),
    mean(draws_syn_vi$beta1),
    mean(draws_syn_vi$`beta_balcony[1]`),
    mean(draws_syn_vi$`beta_balcony[2]`),
    mean(draws_syn_vi$`beta_balcony[3]`),
    mean(draws_syn_vi$sigma)
  )
)

recovery_table_vi[,-1] <- round(recovery_table_vi[,-1], 4)
recovery_table_vi
```

```
##       Parameter True_vi Estimate_vi
## 1         beta0  -2.380     -1.5534
## 2         beta1   0.960      0.8359
## 3 beta_balcony1  -0.190     -0.2252
## 4 beta_balcony2  -0.130     -0.1154
```

```
## 5 beta_balcony3  -0.012      0.0086
## 6         sigma   0.470      0.4858
```